Maintained reproduction of Coppock, Gross, Porter, Thorson and Wood (2023)

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1 What this repository is

This repository re-runs the analysis behind a published article from its deposited data, in current R, and lays every number the article prints against what that analysis produces. It holds three things: a script that fetches and verifies the authors' replication archive, a rewrite of the analysis using current packages, and a ground truth that records, claim by claim, whether the published number reproduces.

Article	Coppock A, Gross K, Porter E, Thorson E and Wood TJ (2023). "Conceptual Replication of Four Key Findings about Factual Corrections and Misinformation during the 2020 US Election: Evidence from Panel-Survey Experiments." *British Journal of Political Science*. https://doi.org/10.1017/S0007123422000631
Replication archive	https://doi.org/10.7910/DVN/WQXZUP
Pre-analysis plan	https://osf.io/2hnv5

1.1 Folder layout

```
download_original.R    fetches and verifies the deposited archive
run_all.R              runs everything, in order
original/              the deposited archive (not redistributed; fetched on demand)
original_manifest.csv  every deposited file, its size and its published checksum
maintained/            the rewrite; maintained/output/ holds every number it produces
ground_truth/          the extraction, the comparison and the gates
errata.qmd             corrections to the published article
```

1.2 How to reproduce

Download the repository, open `coppock_etal_2023.Rproj`, and run:

```
source("run_all.R")
```

`run_all.R` fetches the deposited archive from Harvard Dataverse, verifies every file against its published checksum and byte size, fits every model, writes every figure and table, runs

the deposited scripts in a scratch copy, rebuilds the ground truth and re-verifies the archive. Nothing writes into `original/`.

2 The paper

Eight panel-survey experiments fielded on Lucid and Mechanical Turk between September and November 2020, with 17,629 respondents across 24 experiments built on 21 distinct PolitiFact fact-checks. Each respondent saw three topics; for each topic they were assigned to pure control, misinformation, or misinformation followed by a fact-check. Belief certainty in the false claim is the primary outcome, measured again after at least a week.

The headline estimates are a meta-analytic misinformation effect of 4.3 points and a correction effect of -10.5 points on the 100-point belief certainty scale.

3 Does the deposited archive run?

The deposit is fourteen scripts, six data files and thirty-four saved model objects, in a directory tree of 63 files. Every script was run in a scratch copy, twice: once as the archive ships, and once stripped to data plus code so that a script which passes only because the deposit ships the object it reads is visible as a failure.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 1: Deposited scripts by pass and outcome.

pass	clean	error
as_shipped	12	2
stripped	11	3

Two scripts fail in every pass, for reasons unrelated to the analysis. `appendix_figure_3.R` asks the graphics device for a font family that is not installed, and `appendix_figure_9.R` reads `data/clean/all_panels_long.rds`, a path the deposit does not contain: its copy of that file sits at `data/all_panels_long.rds`.

Stripping the copy to data plus code is what the second pass adds. The analysis scripts read model objects that only `fit_all_models.R` builds, so the stripped pass runs that script first and the analysis scripts after it. That leaves 3 of 14 scripts failing, against 2 as shipped. Two are the pair that fail regardless. The third is `figure_5.R`, and it is a finding of its own: it reads `fitted_models/meta_trait_dic_w1_all_adj`, a deposited file with no extension, while the

deposit's own code writes that object as `meta_trait_dic_w1_all_adj.rds`, which the archive does not contain. The object survives only as a deposited artifact; nothing in the deposit can regenerate it under the name the figure reads.

The deposit writes nothing except model objects. Grepping the fourteen scripts for uncommented calls that touch a file returns thirty-four `write_rds()` lines, all in `fit_all_models.R`, and nothing else: no `ggsave`, no `write_csv`, no `sink`. Every figure is drawn to the active device and every table is printed to the console as LaTeX. So there is no deposited artifact to diff against, and a checksum result on `original/` is a statement about the download rather than about the analysis.

Running the deposit in place would not be harmless. Measured by modification time in a scratch copy, `fit_all_models.R` overwrites 32 deposited model objects, of which 32 come back with different bytes, and adds 2 more, and the figure scripts leave an `Rplots.pdf` behind.

3.1 Do the deposited objects follow from the deposited data?

Every model the deposit saves was refitted from the survey data and compared against the saved object.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 2: Deposited model objects against a refit from the deposited data.

objects	estimates agree to 1e-9	estimates disagree	cells missing on one side
34	23	11	225

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 3: The objects where the refit and the deposit disagree, and what the deposit's own specification gives.

object	rows_deposited	rows_rewrite	estimate	archive_specification
ft_targets_ates_w1_PID_adj	138	132	28.7025088	0
ates_w1_p3_adj	6	6	12.8732396	0
persistence_meta	30	30	7.4147056	NA
ates_w3_p3_adj	6	6	7.0610233	0
ft_targets_ates_w1_all_adj	46	44	4.6340926	0
ft_targets_meta_w1_PID_adj	6	6	2.9569308	NA

meta_trait_cates_w2_p2_adj	18	18	1.7930355	NA
iv_p3_adj	6	6	1.4617805	0
ft_targets_meta_w1_all_adj	2	2	0.6069452	NA
meta_trait_cates_w1_all_adj	36	36	0.5175079	NA
meta_trait_dic_w1_all_adj	18	18	0.2600495	NA

The `archive_specification` column is the same comparison run under the deposit's own model formulas rather than the corrected ones, and where it returns zero the disagreement above it is entirely the correction. The two thermometer objects are the exception and are discussed under the errata below.

Two properties of the comparison are worth stating because a single summary number hides both. The rewrite reverses the confidence interval the deposit prints backwards for every negated contrast, so a maximum taken over all six numeric columns at once is the width of the widest interval in the object and says nothing about agreement. And a cell one side computes while the other returns as missing is invisible to any difference, which is exactly how the political knowledge estimates move: the two `estimatr` versions disagree about which HC2 standard errors are estimable, so the pooling drops a different study.

4 Errata

Nineteen errors in the published article are corrected in `coppock_etal_2023_errata.pdf`, whose values are computed at render time from this repository's output. **One of them changes a conclusion**: entry 3 withdraws the claim that the effects of fact-checks remain detectable more than two weeks after exposure, for the reason entry 9 gives. Everything else the article concludes stands. In summary, and numbered as the note numbers them: **1.** the reported total sample size is 17,681 where the article's own appendix table sums to 17,629; **2.** the count of experiments in which misinformation significantly reduced accuracy is given as twelve where the deposited estimates give 13; **3.** two different appendix floats are both captioned Table 1; and **4.** the four sentences reporting how much of the correction effect survives to a later wave report a figure of the order a fixed-effect pool gives where the appendix gives a random-effects pool of the same ratios, 48.9 per cent at Wave 2 and 34.3 per cent at Wave 3, and report the Wave 3 effect as detectable when neither it nor its ratio is distinguishable from zero; and **5.** the reference list gives Rathje et al. (2022) a single author where the letter has five.

Entry 12 is the only one no ground truth row reaches. It comes from an audit that sends every printed reference entry whole to Crossref and checks the authoritative record back into it, which flagged three entries here. The other two are false positives: the Axios news item under Fischer (2020) and the Pew report under Walker and Gottfried (2019) were each matched to an unrelated journal article, and both entries are correct as printed.

4.1 Corrections the rewrite makes to the deposited code

Four coding errors in the deposit are corrected in `maintained/`. Each is unambiguous, in the sense that the code plainly does something other than what the surrounding code says it is doing.

The thermometer models are fitted on the wrong sample. The deposit's loop over targets reads

```
filter(all_panels_long, fc == fc, topic_short_2 == topic_short_2)
```

Inside `filter()` the bare names resolve to the columns, so both conditions compare a column with itself and neither subsets anything: every target model is fitted on all eight panels pooled, including respondents who were never shown the claim. The loop extracts `fc` and `topic_short_2` one row at a time in order to subset, so the intent is not in doubt. Applying the subset costs one of the twenty-three target rows, because Panel 3 names Barack Obama as a target and never asked the Barack Obama thermometer; that row is recorded in `maintained/output/fits/ft_targets_unfittable.csv`.

This moves Figure 6. The published overall labels are -0.19 and 0.18; the corrected estimates are -0.74 and 0.8. All eight of the figure's labels change, and with them two sentences of prose: the introduction's "smaller than one-quarter of a point" and the Results section's "smaller than half a point" are both true of the published figure and false of the corrected one.

There is a second thing wrong with Figure 6 and it is not the rewrite's doing. Six of its eight labels are exactly what the deposited objects give. The two Overall labels are not: the deposit returns -0.23 and 0.19 where the page prints -0.19 and 0.18. The figure script reads those two labels straight out of `ft_targets_meta_w1_all_adj.rds` and rounds them to two decimals, so the published pair cannot be recovered from the archive as deposited. The cause is not established.

These stay findings here rather than errata. The published cells reproduce from the deposit exactly, so what is wrong with them is a matter of method rather than of arithmetic, and correcting a column of cells the deposit reproduces exactly is a different act from correcting a misstated number. The substantive claim is untouched either way: a correction effect of 0.8 points on a 100-point feeling thermometer is as close to nothing as 0.18 is.

The adjusted three-wave models regress the wrong outcomes. In the deposit, `ates_w1_p3_adj` and `ates_w2_p3_adj` both regress `outcome_w3`, `ates_w3_p3_adj` regresses `outcome_w2`, and `iv_p3_adj` instruments `outcome_w2`. The unadjusted block immediately above pairs each wave with its own outcome, and so does the covariate-adjusted conditional-effect block below, so this is a copy-and-paste slip. Two published artifacts show it. Appendix Table 7's `w1` row holds the wave 3 estimate and its unlabelled row holds the wave 2 estimate, which is why 7 of its 21 cells reproduce where the other six regression tables reproduce completely. That count asks whether a published value appears anywhere in the corrected table;

asked row by row, which is the question a reader of the page asks, every one of the table's cells is wrong where it stands. And Figure 4's bottom-left panel draws its Wave 1 and Wave 2 points at the same height, because they are the same estimate. This is erratum 18, and the blank Wave column that hides it on the page, which the same block produces by declaring the wave factor with the two-wave tables' levels, is erratum 17.

Every negated confidence interval prints backwards. The deposit's `filter_and_flip()` negates the estimate and both interval bounds for the misinformation contrast without reversing their order, so those cells print as `[upper, lower]`. 144 of the 492 published appendix cells are printed this way. The interval itself is right; only the order is wrong. The rewrite swaps the bounds back.

The significance column of the appendix regression tables is not a comparison of numbers. The deposit's table script reshapes each fit with `reshape2::melt`, and where a fit carries a character column outside `id.vars` every melted value becomes a string. The surviving `if_else(p.value < 0.05, "* ", " ")` then compares two strings, so `"0.0144"` sorts before `"0.05"` and earns a star while `"1.28e-10"` sorts after it and does not. The effect is that the most significant estimates in the table are the ones printed without a marker: the published tables carry 205 stars where 256 cells have $p < 0.05$. It is deterministic and visible on the page once you know to look for it. It was recorded here rather than in the errata while it was read as a property of the presentation rather than of an estimate; that reading did not survive the remastered edition, which had to set these tables and could not set a marker it knew to be on the wrong cell. It is erratum 15, as the transposed intervals above are erratum 14.

4.2 The persistence percentages are pooled one way in the main text and another in the appendix

The article states three times that fact-check effects persist "at 66 per cent" of their original magnitude after one week, once at the sharper "66.4 per cent", and "at 50 per cent" after more than two weeks. The appendix reports the same quantity and gives different numbers.

The quantity the pre-analysis plan defines, and the one the appendix's own persistence tables print in their `ratio` rows, is the per-experiment ratio of the later-wave coefficient to the Wave 1 coefficient, pooled across experiments by meta-analysis. Pooled with random effects, which is what `metafor` does by default and what the appendix's meta-analysis rows carry, it is 48.9 per cent at Wave 2 and 34.3 per cent at Wave 3 with covariates, and 54.4 and 38.9 per cent without them. Pooled with fixed effects, the same ratios give 64.9 and 70.8 per cent at Wave 2 and 52.2 and 63.3 per cent at Wave 3. The ratio of the pooled Wave 2 effect to the pooled Wave 1 effect, a different derivation of the same idea, gives 46.9 per cent, and the corresponding Wave 3 figure is 39.7 per cent.

The published percentages are of the order the fixed-effect pool gives and not of the order the appendix's own rows give, so a reader of the introduction takes away a materially different number from a reader of the appendix. No combination of adjustment, estimator and sample

returns 66.4 or 50 exactly. The article should have used random effects throughout, and the four sentences carrying these figures are corrected on that basis in `errata.qmd`. The Wave 3 correction is larger than the pooling alone would make it, because the ratio behind appendix Table 7 also instruments the wrong outcome; see the three-wave finding above.

4.3 Two further quantities with no counterpart

The appendix states that attitudinal outcomes were observed for 22 political figures and groups. The deposited data carry 14 feeling thermometer columns. And appendix Table 9 reports a mean adjusted r-squared for each of ten covariates across "our 21 experiments, resulting in 210 separate linear models"; the deposit ships no code for that table, and the article elsewhere describes 24 experiments. The rewrite adds no script for it, since writing one would be estimating something the deposited archive never did.

5 The ground truth

`ground_truth/coppock_etal_2023_ground_truth.csv` carries one row per published claim. `value_paper` is the string the article prints and is the only column anyone typed; `value_script` is read out of the deposit and `value_rewrite` out of `maintained/output/`, both by script. A value agrees when the computed number, printed to the page's own precision, gives the same digits.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 4: Published claims by type and verdict.

claim_type	does not reproduce	no verdict	reproduces	does not hold	holds
definitional	3	9	21	0	0
descriptive	0	2	0	3	15
pipeline	23	11	128	0	0
structural	1	12	12	0	0
transcribed	0	14	0	0	0

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 5: Where the fault lies, on every row that is not a clean match.

defect_locus	rows
archive	31
paper_internal	8
unresolved	2

Every row that is not a clean match carries a `defect_locus`, and no clean match carries one. The five values are `paper_internal` (the article disagrees with its own tables or its own data), `archive` (the deposit cannot support the claim), `environment` (R or a package moved underneath it), `rewrite` (ours) and `unresolved` (cause not established).

5.1 Coverage, float by float

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 6: Coverage per published float. Printed numbers are what a float puts on
and covered counts those laid against an instrument; plotted is what a fi

float	printed numbers	covered	reproduced by rewrite	reproduced by
Appendix Figure 1	0	0	0	
Appendix Figure 2	0	0	0	
Appendix Figure 3	0	0	0	
Appendix Figure 7	0	0	0	
Appendix Figure 8	0	0	0	
Appendix Figure 9	12	0	0	
Appendix Figures 4 to 6	0	0	0	
Appendix regression table 1	400	400	400	
Appendix regression table 2	400	400	400	
Appendix regression table 3	400	400	400	
Appendix regression table 4	300	300	300	
Appendix regression table 5	300	300	300	
Appendix regression table 6	84	84	84	
Appendix regression table 7	84	84	28	
Appendix Table 1 (sample composition)	206	9	9	
Appendix Table 8	72	0	0	
Appendix Table 9	10	0	0	
Figure 1	0	0	0	
Figure 2	72	72	72	
Figure 3	68	0	0	

Figure 4	0	0	0
Figure 5	36	36	31
Figure 6	8	8	0

Seven of the appendix's floats and two of the article's print no numbers at all. For those the checkable published quantity is the count of what they plot, and that count is verified; a table plots nothing, so its plotted columns are left empty rather than reported as zero.

Two floats are covered thinly and both are worth stating plainly. **Figure 3 is the largest gap in this repository.** It labels every effect significantly different from zero and every one of its eight meta-analytic summaries, roughly 68 printed numbers, and none of those labels was transcribed. What is verified is that the figure plots 96 conditional effect estimates, and that those estimates are the covariate-adjusted cells of appendix Tables 2 and 3, every one of which reproduces. **Appendix Table 1** prints 206 numbers, of which the eight panel sizes are checked individually and the rest are checked as counts of non-empty cells.

The seven appendix regression tables are the bulk of the published record: 492 cells, each printing an estimate, a standard error and two interval bounds, so 1968 published numbers. All of them are parsed positionally from the published PDF into `ground_truth/published_appendix_values.csv` and compared cell by cell.

6 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: 254 rows, one per numeric claim in the article and its appendix, each classified by hand as `pipeline`, `descriptive`, `definitional`, `structural` or `transcribed`, and each carrying the string the page prints and the precision it prints it at. Spelled-out numbers were swept for separately from digits, since no token scan sees "eight panel experiments", "twelve out of the twenty-four opportunities" or "thirteen days"; 20 of the extraction's quoted sentences state their number in words.

Two instruments read the same pipeline output by separate paths. `ground_truth/build_ground_truth.R` builds the comparison table. `maintained/in_text_claims.R` carries the article's own sentence in a block comment beside code that recomputes the number, and prints 219 claims in the form `CLAIM <id> = <value> || [verdict] <label>`. It reads `maintained/output/` and the extraction, never the ground truth, so the two derivations are independent and a disagreement between them is a finding. The verdict comes from `excheckr`, which derives it from the extraction and the errata spine: a claim matches the page, or a quantity erratum names it and it matches the corrected value instead, or its block declares that it computes something other than the printed token (`derived`) or that the deposit has no counterpart (`absent`), or it is a descriptive claim that disagrees, scored `hedged` with its verdict left to the ground truth's `holds`. Any other outcome stops the run, and the file's last line counts every verdict.

The build asserts all of the following and stops if any fails.

- Every `pipeline` and `descriptive` claim has a block and a row.
- The number of printed claims equals the number of rows requiring one, and the two sets of ids are equal in both directions.
- Every value the two instruments both produce agrees exactly at the article's precision.
- The claims the extraction's `expect` column marks `derived` or `absent` are exactly those the claims file declares so.
- Re-rendering `value_paper` from its own number at its own recorded precision returns the string the extraction stores, which is what catches a precision that is right about the value and wrong about the digits.
- Every adverse row carries a `defect_locus` and no clean match does.
- Every `paper_internal` row carries a `disposition`, which is `corrected_here` wherever the errata spine names the claim.
- Every published float has at least one row.

Each of the first, third and fourth was tested by breaking it: deleting a block fails the count, perturbing a printed value fails the cross-check, and changing one claim's recorded precision fails the string check.

7 The maintained rewrite

`maintained/` is 22 scripts. `fit_all_models.R` fits every model from the deposited survey data and writes each family as a tidy CSV under `maintained/output/fits/`, so every number a figure or table prints can be traced to a file a reader can open. Nothing downstream reads the deposit's saved objects.

The substitutions the rewrite makes are the usual ones for code of this age: the native pipe for `magrittr`, `pivot_longer`/`pivot_wider` for `reshape2::melt`/`dcast`, `reframe(tidy(...))` for `do(tidy(...))`, `pick(everything())` for `cur_data()`, `write_csv()` for `xtable` printed to the console, `linewidth` for the deprecated `size` aesthetic, and `position_dodge()` for `ggstance::position_dodgev()`. The Roboto font dependency in the traffic figure is dropped. `plyr` is not loaded at all, which removes the masking hazard the deposit works around by loading it first.

8 Figure verification

Every rendered figure was laid beside the published one. Numbers alone cannot catch a transposed axis or a mislabelled series, and three of this article's six figures print their estimates on the face of the plot, which makes them published tables in disguise.

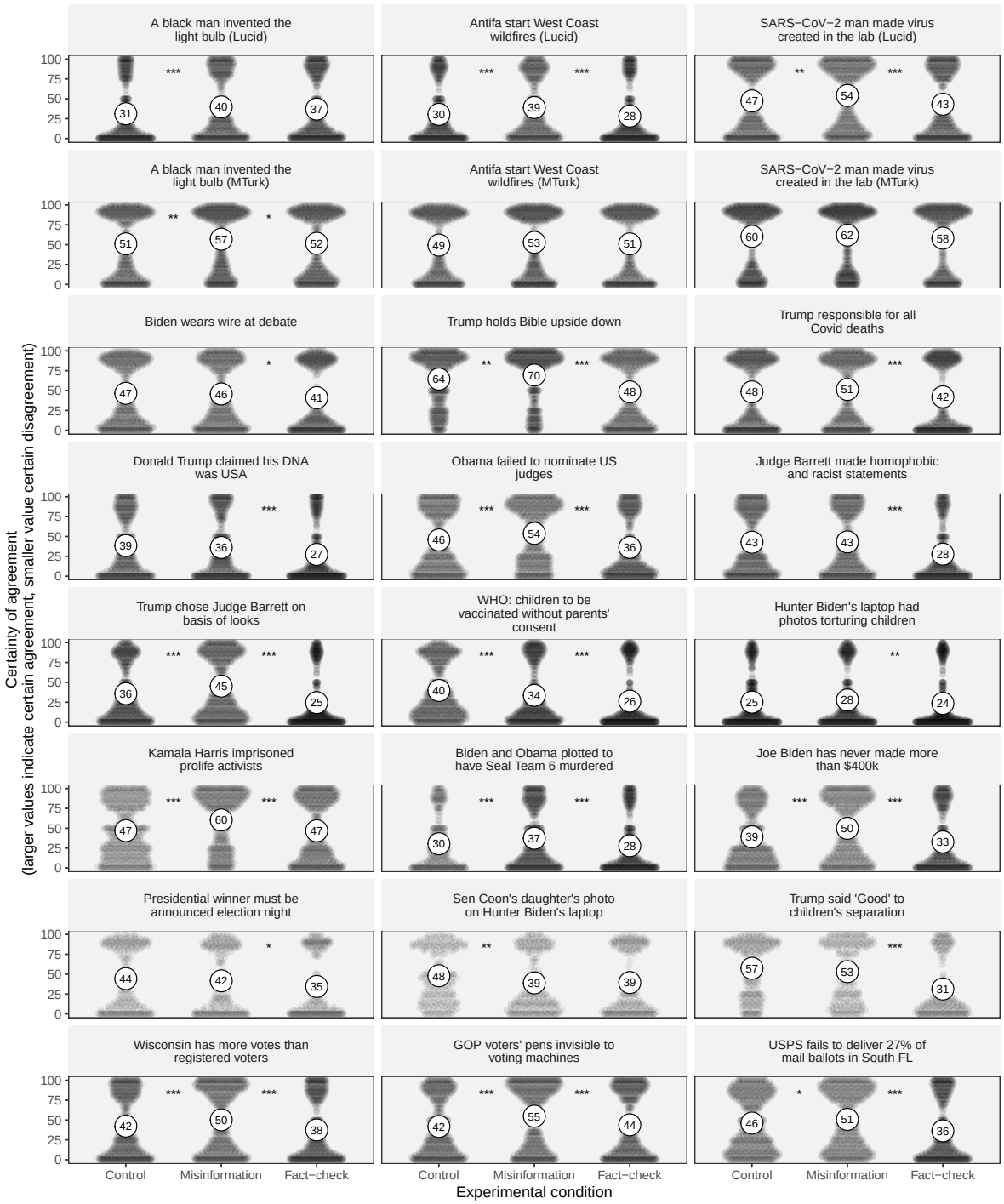


Figure 1: Figure 2 as the rewrite draws it. All seventy-two conditional means reproduce the published labels exactly.

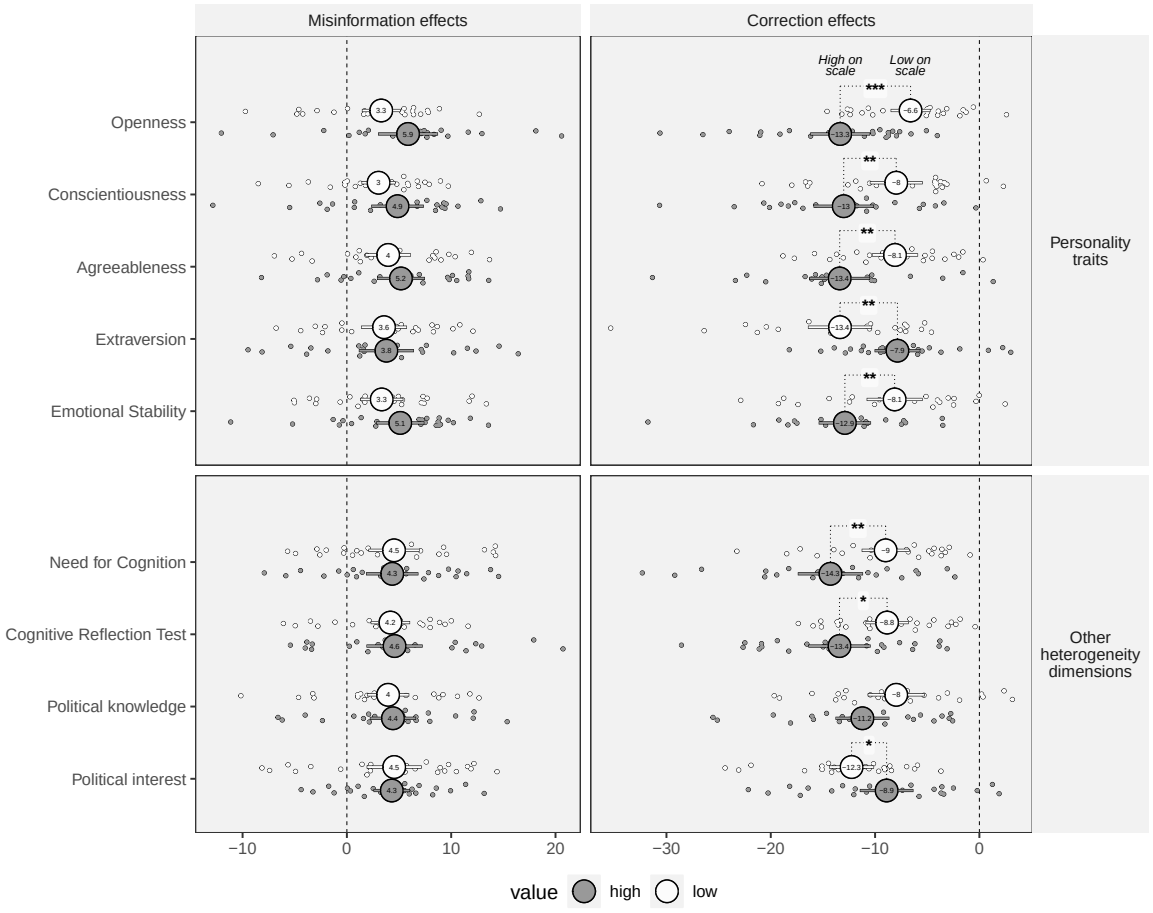


Figure 2: Figure 5 as the rewrite draws it. Thirty-one of the thirty-six labels reproduce the published ones; the five that move are errata entry 10, where the deposit's pooling dropped an experiment whose standard errors it could not compute.

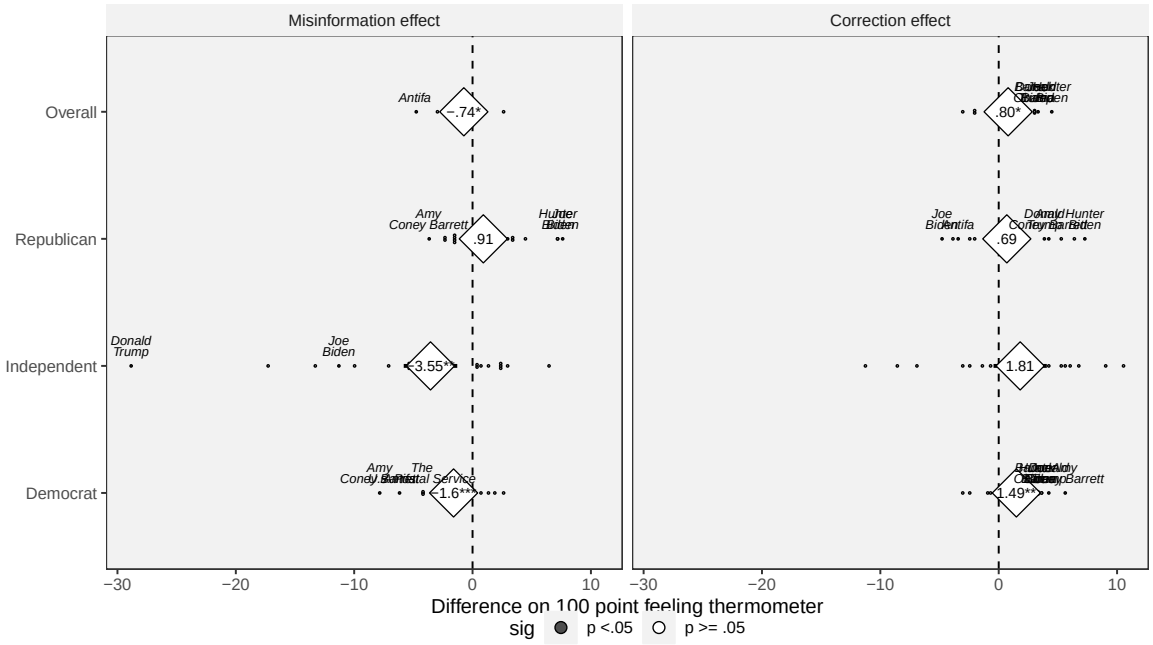


Figure 3: Figure 6 as the rewrite draws it, with the thermometer models fitted on the panel that carried each claim. All eight labels differ from the published ones.

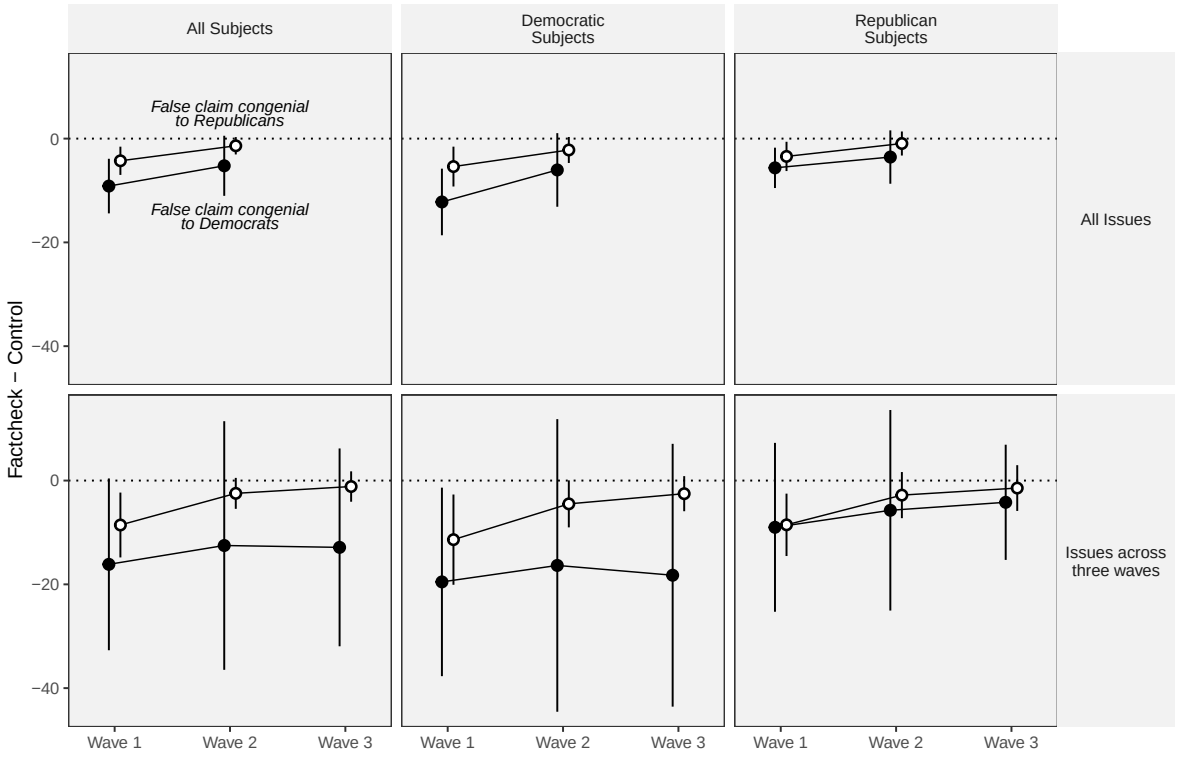


Figure 4: Figure 4 as the rewrite draws it. In the published version the Wave 1 and Wave 2 points of the bottom-left panel sit at the same height, because the deposit fits both on the Wave 3 outcome.

9 Rewrite verification

`run_all.R` was run to completion twice from clean sessions and the whole tree diffed. Every CSV is byte-identical between runs. The figure PDFs differ, because a PDF records the time it was written.

`original/` is verified twice per run: `download_original.R` is sourced first, which checks every file against the MD5 Dataverse serves for `?format=original` and against its deposited byte size, and refuses to continue if `original/` holds any file the manifest does not list, dotfiles included. It is sourced again as the last step, because the first pass proves only that the archive was intact when the run began. All 63 files match, and 0 carry a published checksum that disagrees with the bytes Dataverse serves. Checksums for files this repository writes are not quoted anywhere: a PDF records its write time, so its hash changes on every run.

10 R environment

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

component	version
R	4.6.2
tidyverse	2.0.0
estimatr	2.0.1
metafor	5.2.1
ggplot2	4.0.3
broom	1.0.13

The rewrite keeps the current sampler and the current packages throughout. Nothing in this analysis draws at random: no script calls `sample()`, `rnorm()` or a bootstrap, so the results are bit-identical at any seed and the sampler change in R 3.6 is not in play.